Four approaches to a tensor-diagram library

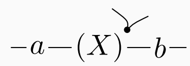
Same test diagrams in each approach. Shared base: `tn` picture style, `copy`, `flat left/right` node styles.

A. Legs as node options

(plain tikz + two new keys)

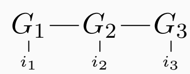
A1: derivative of $a^T X b$

```
\begin{tikzpicture}[tn]
  \node[legs={180}] (a) {$a$};
  \node[right=.8em of a, legs={135*,45*}]
      (X) {$(X)$};
  \node[right=.8em of X, legs={0}]   (b) {
      $b$};
  \draw (a) -- (X) -- (b);
\end{tikzpicture}
```



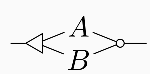
A2: TT chain with labeled legs

```
\begin{tikzpicture}[tn]
  \node[legs={270/i_1}] (G1) {$G_1$};
  \node[right=1em of G1, legs={270/i_2}] (
      G2) {$G_2$};
  \node[right=1em of G2, legs={270/i_3}] (
      G3) {$G_3$};
  \draw (G1) -- (G2) -- (G3);
\end{tikzpicture}
```



A3: Khatri–Rao $A * B$

```
\begin{tikzpicture}[tn]
  \node[flat left, legs={180}] (f) {};
  \node (A) at (.55,.22)  {$A$};
  \node (B) at (.55,-.22) {$B$};
  \node[copy, legs={0}] (c) at (1.1,0) {};
  \draw (f) -- (A) -- (c);
  \draw (f) -- (B) -- (c);
\end{tikzpicture}
```

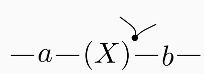


B. The graphs library

(chains as one expression)

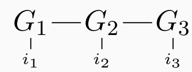
B1: derivative of $a^T X b$

```
\begin{tikzpicture}[tn]
\graph[grow right sep=.8em] {
  l[as=, coordinate] -- a[as=$a$] --
  X[as=$(X)$, legs={135*,45*}] --
  b[as=$b$] -- r[as=, coordinate]
};
\end{tikzpicture}
```



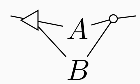
B2: TT chain

```
\begin{tikzpicture}[tn]
\graph[grow right sep=1em] {
  G1[as=$G_1$, legs={270/i_1}] --
  G2[as=$G_2$, legs={270/i_2}] --
  G3[as=$G_3$, legs={270/i_3}]
};
\end{tikzpicture}
```



B3: Khatri–Rao (fan-out/fan-in)

```
\begin{tikzpicture}[tn]
\graph[grow right sep=.6em,
       branch down sep=.5em] {
  l[as=, coordinate] --
  f[as=, flat left] --
  { A[as=$A$], B[as=$B$] } --
  c[as=, copy] -- r[as=, coordinate]
};
\end{tikzpicture}
```

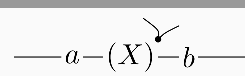


C. A mini-DSL

(dashes are edges, like the book's inline notation)

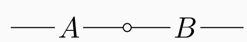
C1: derivative of $a^T Xb$

```
\tn{-a-{X'}-b-}
```



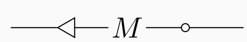
C2: matrix chain with copy dot

```
\tn{-A-*-B-}
```



C3: Khatri–Rao-ish chain

```
\tn{-<-M-*-}
```

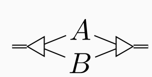


D. Pics: reusable molecules

(with anchors, so they compose)

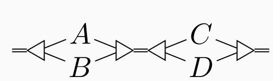
D1: Kronecker bundle

```
\begin{tikzpicture}[tn]
  \pic (k) {kron={A}{B}};
  \draw[double] (k-in) -- ++(-.35,0);
  \draw[double] (k-out) -- ++(.35,0);
\end{tikzpicture}
```



D2: mixed product (511) by composition

```
\begin{tikzpicture}[tn]
  \pic (k1) {kron={A}{B}};
  \pic (k2) at (1.6,0) {kron={C}{D}};
  \draw[double] (k1-in) -- ++(-.35,0);
  \draw[double] (k1-out) -- (k2-in);
  \draw[double] (k2-out) -- ++(.35,0);
\end{tikzpicture}
```



D3: determinant stack + TT chain

```
\begin{tikzpicture}[tn]
  \pic {det stack=X};
\end{tikzpicture}
\qquad
\begin{tikzpicture}[tn]
  \pic {tt chain={G_1,G_2,G_3}};
\end{tikzpicture}
```